

MGT 413 Operations Management

Winter 2026

Prof. Robert Montgomery

Email: romontgomery@ucsd.edu

Teaching Assistant: Siqi Wang

Email: siw119@ucsd.edu

Course Location: OTRSN 4E106, Wednesdays, 11:00am-1:50pm.

DESCRIPTION

Businesses of any type or size can benefit from careful analysis of the processes that comprise their operations. This course will focus on how to understand and present a business as a series of processes and how to analyze these processes to improve performance. Students will be introduced to the foundations of Operations Management, with emphasis on identifying how businesses structure operations to align with their strategic goals. Through consideration of a range of businesses, we will identify how each makes decisions to manage operations and gain the mathematical skills needed to analyze and improve these decisions.

OBJECTIVES

At the close of this course, you will be able to:

- Identify the tradeoffs companies make between the four product attributes on which firms compete: cost, quality, variety, and response time, for a segment of customers
- Understand why businesses make certain operational decisions to deliver on the attributes they have chosen to align with their strategic position
- Create process flow diagrams for company processes in order to conduct analysis
- Use Little's Law to relate flow times, throughput, and inventory in a stable process
- Examine the factors affecting flow time and identify ways to reduce flow time within a process
- Conduct capacity analysis to identify the bottlenecks and utilization rates of resources
- Explain and optimize different inventory management strategies
- Present an in-depth assessment of a single business's operations that you select for your midterm project

MATERIALS

Required

- *Managing Business Process Flows: Principles of Operations Management* by Anupindi et al., 3rd Edition, 2012, Prentice Hall, ISBN: 0136036376
- Course Reader:
 - a. Case- Shouldice Hospital
 - b. Deep Change
 - c. Case- State Automobile License Renewal
 - d. Case- Benihana of Tokyo
 - e. Case- National Cranberry Cooperative
 - f. Littlefield Technologies
 - g. Case- Toyota Motor Manufacturing, U.S.A., Inc.
 - h. Bullwhip Effect in Supply Chains

SCHEDULE

Date	Class Topic	Class Prep work
Week 1 January 7	Introduction to Operations Management	-Shouldice Hospital -Deep Change -Textbook Chapters 1 and 2
Week 2 January 14	Process Flows & Little's Law	-State Automobile License Renewal -Textbook Chapter 3
Week 3 January 21	Flow Time Analysis	-Textbook Chapter 4
Week 4 January 28	Capacity Analysis	-Benihana of Tokyo -Textbook Chapter 5
Week 5 February 4	National Cranberry Case & Littlefield Technologies	-National Cranberry Case -Littlefield Technologies
Week 6 February 11	Inventory Management: Part 1	-Midterm Project due (8pm the night before class February 10) -Textbook Chapter 6
Week 7 February 18	Inventory Management: Part 2 House Build Game	-Textbook Chapter 7
Week 8 February 25	Lean Operations	-Toyota Motor Manufacturing -Textbook Chapter 10
Week 9 March 4	Beer Game	-Textbook Chapters 8 & 9 -Beer Game Prep Work
Week 10 March 11	Supply Chain Management and Course Review	-Bullwhip Effect in Supply Chains

Assignments

To review the material covered in class and help prepare for the weekly quizzes, there will be weekly homework assignments. You are free to work with classmates on these assignments. The assignments will allow for multiple attempts. These assignments are designed to guide your learning more so than to assess your understanding of the material

The class also includes a midterm project that will require you to analyze a company of your choosing. The report will be due in Week 6. Late submissions for this assignment will still be graded, but will carry deductions commensurate with the delays in submission. The project will focus on your ability to map a company's processes for deeper operations analysis. Details for the midterm project will be provided in class and on Canvas.

As part of homework and class participation assessments, the professor will occasionally ask students to prepare additional analysis of the in-class discussions and games.

Quizzes

This class will include weekly quizzes, which will occur at the start of each class to ensure students have reviewed the material prior to class. These quizzes will also allow the professor to understand which topics require additional attention. The two lowest quiz grades in the quarter will be dropped. This will also allow students to miss up to two classes without penalty. The professor will provide an overview before each class addressing which material is fair game for the upcoming quiz.

Reading Assignments

Weekly readings will follow the *Managing Business Process Flows: Principles of Operations Management* textbook. You will also be expected to read the assigned material (primarily cases) from the course reader to prepare for class.

Final Exam

The final exam will take place on Wednesday, March 18, 11:30am-2:30pm in OTRSN 4E106.

GRADING

Assignments	Percentage
Class Participation	20%
Quizzes	20%
Homework Assignment	10%
Midterm project	25%
Final Exam	25%
Total	100%

CHATGPT AND OTHER AI TOOLS

ChatGPT and other AI tools are now widely available. In your careers you will have to figure out when these tools are helpful and when they are not. This course is designed to assess students primarily in the classroom setting. I encourage you to use AI tools in any way that will help guide your learning

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of university intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:
<http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2>

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.